AI Based Smart Retail Recommendation System

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Project Proposal Report

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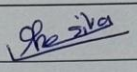
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The retail industry is moving toward more personalized shopping experiences, but many existing solutions still do not fully connect digital features with real in-store shopping. Smart mirrors are used in some stores for virtual try-ons, but they are expensive, fixed in one place, and offer limited intelligent recommendations. There is a clear research gap in providing a flexible, mobile-based solution that can deliver personalized recommendations while understanding customer behavior in real time.

This research proposes a **mobile application-based smart retail recommendation system** using **Augmented Reality (AR)** and **Virtual Reality (VR)** technologies. The system is developed using **Unity**, allowing customers to try on clothes virtually through their mobile devices without needing a physical smart mirror. The app creates an interactive and immersive shopping experience where users can view products on themselves in real time.

In addition, the mobile application includes a real-time **emotion detection feature**. Using the device camera, the system analyzes facial expressions to understand customer emotions while trying different outfits. Techniques such as computer vision and deep learning models (e.g., CNN) are used to detect emotional responses like satisfaction or dissatisfaction. Based on these emotions and user interactions, the system provides more accurate and personalized product recommendations.

The expected outcome is a smart and user-friendly shopping experience where customers can explore, try, and select products anytime and anywhere. This approach reduces the need for expensive in-store hardware while improving personalization and customer engagement. The system also considers user privacy by processing sensitive data securely and minimizing data sharing.

Keywords: Mobile App, Augmented Reality, Virtual Reality, Emotion Detection, Personalized Retail, Unity

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1. INTRODUCTION

1.1 Background and Context

The retail industry is increasingly adopting digital technologies to improve customer experience and make shopping more convenient. Traditional retail stores often provide limited personalized support when customers are selecting clothing or fashion products. As a result, many customers find it difficult to decide which items suit them best. Technologies such as Artificial Intelligence (AI), Augmented Reality (AR), and recommendation systems are now widely used to address this issue and support better purchasing decisions [1].

In recent years, AR and Virtual Reality (VR) technologies have enabled virtual try-on experiences, allowing customers to visualize clothing without physically wearing them. Unlike traditional smart mirror systems that are fixed in stores, mobile applications provide a more flexible and accessible solution. Using platforms like Unity, AR-based mobile apps allow users to try on outfits in real time through their smartphones, making the shopping experience more interactive, immersive, and convenient [2].

However, most existing solutions mainly focus on visualization and do not provide intelligent recommendations based on user behavior or emotional responses. They often fail to analyze important factors such as customer emotions, preferences, and real-time reactions while trying products. Research shows that emotion-aware systems using computer vision and deep learning techniques can better understand user satisfaction and improve recommendation accuracy [3], [5].

To address this research gap, this project proposes a **mobile application-based smart retail recommendation system** that integrates AR/VR technologies with real-time emotion detection. The system uses the mobile device camera to analyze facial expressions and identify customer emotions while interacting with virtual clothing. Based on these emotional responses and user behavior, the system provides personalized product recommendations. This approach creates a more intelligent and user-friendly shopping experience while reducing the need for expensive in-store hardware.

Overall, the proposed system supports the advancement of digital retail by combining mobility, immersive technologies, and AI-driven personalization. It contributes to building a smarter retail environment that improves customer satisfaction and engagement while aligning with modern technological trends [4], [6].

1.2 Literature Review

Recent research shows that digital technologies such as artificial intelligence (AI), augmented reality (AR), virtual reality (VR), and computer vision are increasingly used in the retail industry to improve the customer shopping experience. Many studies have explored the use of mobile-based virtual try-on systems in fashion retail. These systems allow customers to see how clothes look on them without physically wearing them by using smartphone cameras and AR technology. This makes the shopping process faster, more flexible, and interactive while also allowing retailers to provide digital experiences beyond physical stores [1]. Research also shows that virtual try-on technology can reduce the need for physical fitting rooms and improve customer satisfaction by allowing users to quickly test different clothing styles and colors through mobile devices [2].

Several studies also discuss the use of AI-based fashion recommendation systems. These systems analyze factors such as clothing style, color, body shape, and customer preferences to recommend suitable outfits [3]. Some research focuses on emotion-aware recommendation systems that analyze facial expressions and behavioral signals to understand customer reactions during shopping using mobile cameras [4]. Other studies highlight the use of computer vision and deep learning techniques to detect facial features and user preferences in order to generate personalized fashion suggestions [5]. In addition, context-aware recommendation systems have been proposed to provide suggestions based on user behavior, interaction patterns, and real-time conditions during the shopping process [6]. These studies show that intelligent recommendation technologies can significantly improve product discovery and customer engagement in digital retail environments.

Different research approaches have been used to develop smart retail recommendation systems. Many systems use computer vision models to detect facial expressions, user posture, and interaction patterns from camera data [7]. Deep learning algorithms are also used to analyze visual inputs and identify emotional responses and preference patterns. Some studies combine multiple data sources such as facial expressions, user interactions, and product metadata to generate more accurate and personalized recommendations [8]. These approaches aim to create intelligent shopping experiences by understanding the customer's emotions and behavior in real time.

In addition to research studies, several mobile applications and AR-based retail solutions have already been introduced. Modern systems provide virtual try-on features, interactive user interfaces, and immersive experiences using AR and VR technologies developed through platforms like Unity [9]. Some applications also integrate product catalogs and online inventory systems to display details such as size, price, and availability. These technologies help users explore products easily and enhance engagement through digital interaction.

However, existing systems still have several limitations. Most current solutions mainly focus on virtual try-on and visualization features. They often do not provide intelligent product recommendations based on customer emotions or real-time behavior. Many systems also lack personalization and do not analyze how users react while interacting with products. Another limitation is that some applications are not effectively connected with real-time inventory data, which means recommended items may not always be available.

Therefore, there is a need for a more advanced mobile-based retail recommendation system. Current AR and VR applications mainly focus on visualizing clothing but do not provide smart recommendations. They also lack emotion-based personalization, real-time interaction analysis, and efficient integration with product availability. The proposed system addresses this gap by using AI-driven emotion detection and AR-based interaction to recommend suitable products through a mobile application. This approach aims to improve the shopping experience and support more intelligent and personalized retail recommendation systems.

1.3 Research Gap

Mobile-based augmented reality (AR) and virtual reality (VR) technologies are increasingly used in modern retail to improve the shopping experience. These technologies allow customers to try on clothes virtually using smartphone cameras without physically wearing them. This makes shopping more flexible and convenient, as users can access these features anytime and anywhere [1]. At the same time, AI-based fashion recommendation systems have been developed to suggest outfits based on factors such as clothing style, color, and customer preferences [2]. Researchers have also explored emotion-aware recommendation systems that analyze facial expressions, user behavior, and other contextual information to understand customer reactions and improve product suggestions in real time [3]. In addition, AI-based retail analytics can analyze user interactions within mobile applications to help understand customer behavior and improve recommendation strategies [4].

However, many existing mobile AR/VR retail applications still have several limitations. Most systems mainly focus on virtual try-on and visualization features and do not provide intelligent product recommendations. They also do not effectively analyze important factors such as customer emotions, behavior, or interaction patterns while suggesting products [1]. Some AI-based fashion recommendation systems operate independently and are not integrated with real-time AR interaction within mobile applications [2]. Another major limitation is the lack of personalization and real-time data integration. Many systems recommend products without considering dynamic user responses or real-time product availability [3]. Because of these limitations, there is a need for an advanced mobile-based retail system that can recommend suitable products using AI based on real-time user interaction and emotional responses. The proposed system aims to address this gap by combining AR/VR technology, emotion detection, and behavioral analysis within a mobile application to provide more personalized and intelligent product recommendations [4].

Feature	[1] Smart Mirror AR Retail Study	[2] AI Fashion Recommendation System	[3] Context-Aware Emotion Recommendation	[4] AI Consumer Response Analysis
Virtual Try-On	✓	✕	✕	✕
AI Recommendation	Limited	✓	✓	Limited
Emotion / Behavior Analysis	✕	✕	✓	✓
Real-Time Inventory Integration	✕	✕	✕	✕
Smart Mirror Interaction	✓	✕	✕	✕
Personalized Product Suggestions	Limited	✓	✓	Limited

Table 1 -

2. OBJECTIVE

2.1 Main Objective

This project develops an AI-powered mobile application for retail that uses augmented reality (AR) and virtual reality (VR) to analyze customer facial expressions, interaction behavior, and emotional responses in real time using computer vision and recommends suitable clothing products, improving personalization and the overall smart retail shopping experience.

2.2 Specific Objective

1. To design and develop a mobile application interface using AR/VR that allows customers to view clothing items virtually and interact with the system while shopping.
2. To implement computer vision techniques to detect customer facial features, emotional expressions, and interaction behavior in order to understand customer preferences.
3. To develop an AI-based recommendation module that suggests suitable clothing products based on customer emotions, features, and behavior.
4. To integrate the mobile application with the product database so that the system recommends only available products in real time.
5. To improve personalized shopping experience by providing relevant clothing suggestions through the mobile AR/VR-based system.

3. METHODOLODGY

The methodology for this component follows a Design Science research approach because the main goal is to design and develop a technological solution for improving the retail shopping experience. The system focuses on creating a mobile application-based retail recommendation system that can suggest suitable clothing products to customers while they interact with the application using AR/VR features. The development process includes designing the system, implementing the required technologies, and testing the system to evaluate how effectively it provides personalized recommendations. The main technical methods used in this system include computer vision and artificial intelligence techniques to analyze customer emotions and interaction behavior in real time.

The system will use several technical methods and development tools to build the mobile AR/VR solution.

The important methods and tools used are listed below:

- Computer Vision and AI Algorithms – Used to detect customer facial features, emotional expressions, and interaction behavior through the mobile device camera.
- Augmented Reality and Virtual Reality (AR/VR) – Used to provide virtual try-on functionality and create an interactive shopping experience within the mobile application.
- Recommendation Algorithm – Used to suggest suitable clothing items based on customer emotions, features, and preferences.
- Unity Development Platform – Used to develop the AR/VR environment and interactive user interface of the mobile application.
- Programming Languages (C# and Python) – C# is used within Unity for application development, while Python supports AI and machine learning model development.
- OpenCV Library – Used for image processing and feature detection from camera input.
- TensorFlow or PyTorch – Used to develop deep learning models such as CNN for emotion detection and feature analysis.
- Database System (Firebase or Cloud Database) – Used to store product details and application data.

The system will collect limited data such as facial images and user interaction behavior only for system analysis and testing. All collected data will follow ethical and privacy guidelines, and sensitive data will not be stored permanently. To evaluate the system, several validation methods will be used. These include:

- Recommendation Accuracy – Measuring how correctly the system suggests suitable clothing items based on user interaction and emotions.

• User Feedback – Collecting user opinions about the usefulness and usability of the application.

• System Performance – Measuring how well the application responds during real-time interaction and AR processing.

These evaluation methods will help determine whether the mobile AR/VR-based system improves personalization and enhances the overall retail shopping experience.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and its Components

The proposed system is a **Smart Retail Recommendation System** designed to improve customer experience and support intelligent decision-making in retail stores. The system uses artificial intelligence, computer vision, and data analytics to understand customer behavior, optimize inventory management, and provide personalized recommendations and marketing strategies.

The overall system is composed of **four main components** that work together to create an intelligent retail environment:

- **Customer Emotions and Behavior Detection** – This component uses cameras and AI to detect customers entering the store and analyze their emotions and movements. It identifies facial expressions and shopping behavior to understand customer reactions and product interests.
- **Mobile AR/VR Recommendation System –** This component provides an interactive AR/VR interface developed in Unity that allows customers to try products virtually and receive personalized recommendations based on real-time emotional and behavioral analysis.
- **Smart Inventory and Stock Flow Optimization** – This component manages inventory levels across stores and warehouses. It predicts product demand and optimizes stock movement to ensure products are available when needed.
- **Personalized Marketing Intelligence** – This component analyzes customer purchasing behavior and market trends to generate targeted marketing promotions and personalized campaigns.

These components work together to create a connected retail system where customer behavior, product availability, and marketing strategies are continuously analyzed and optimized.

Figure 2: System Overview Diagram

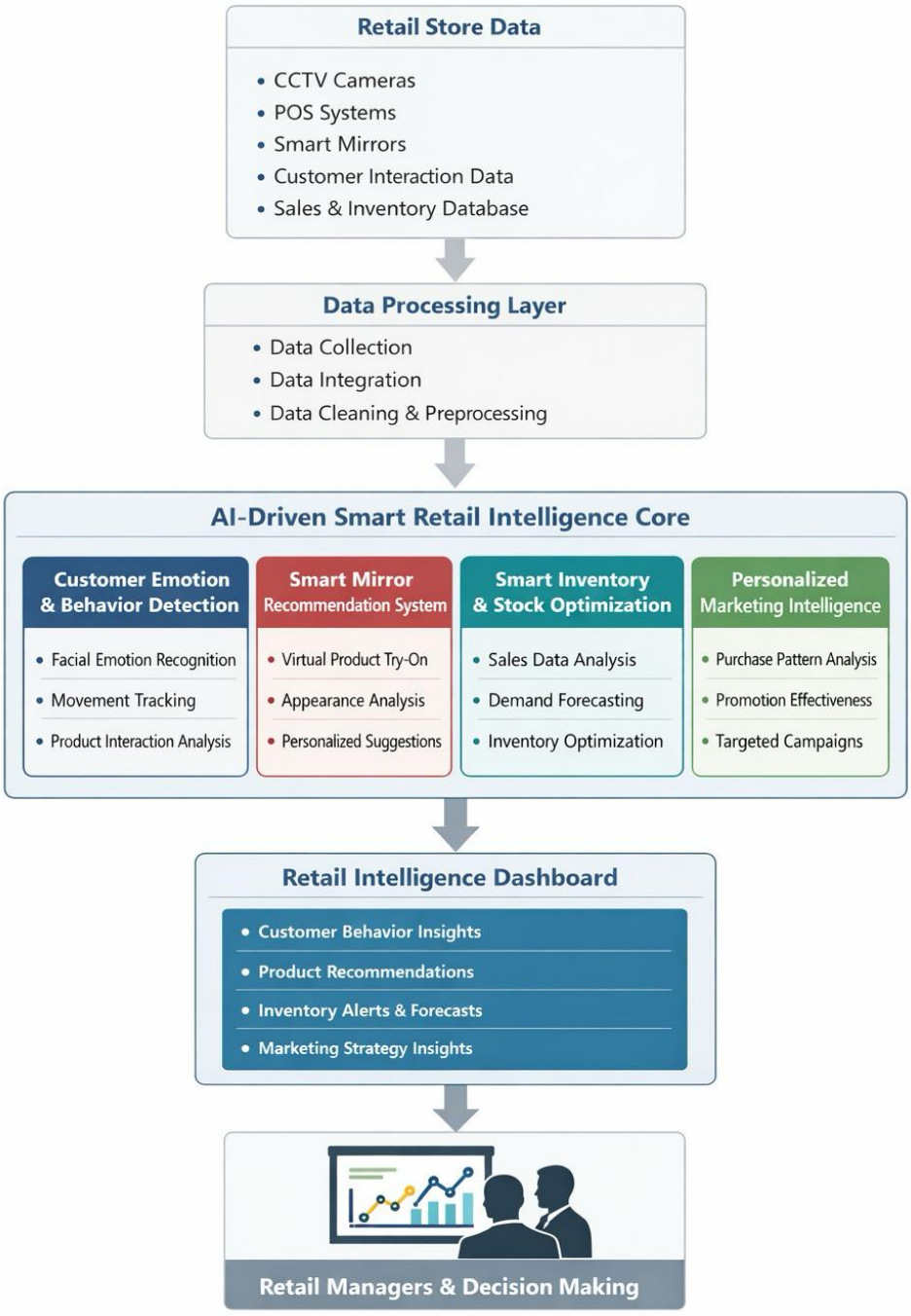
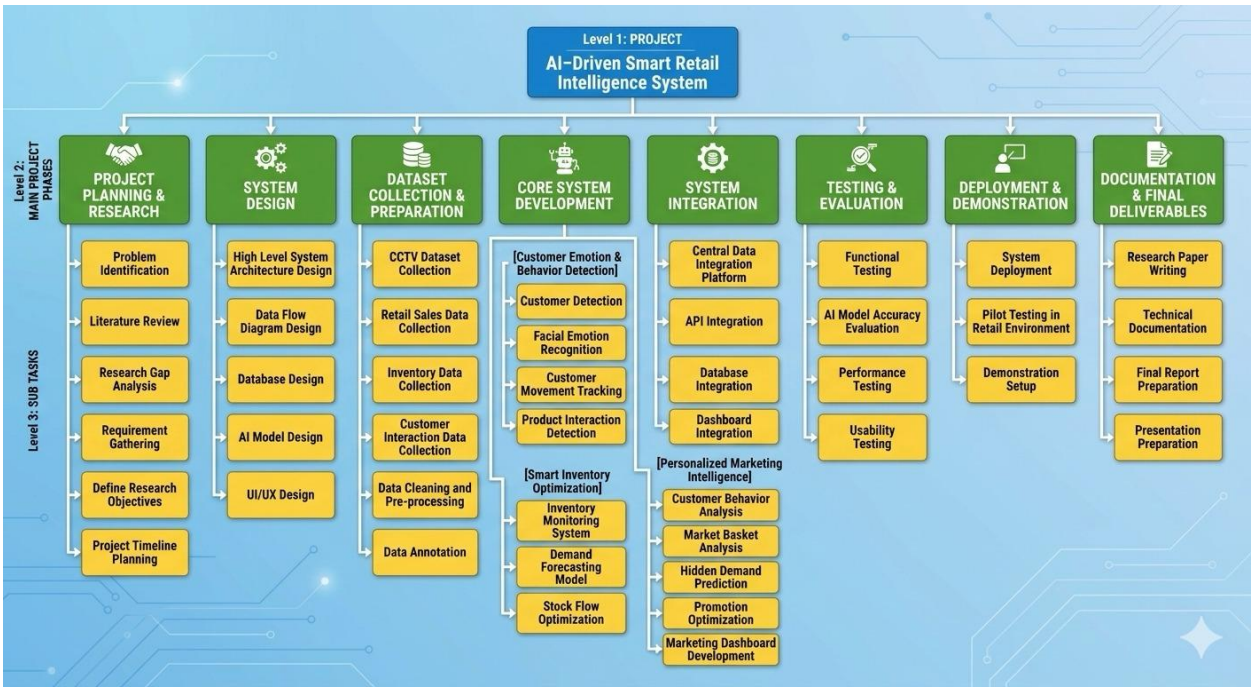


Figure 1. Data Flow Diagram



4.2 Component Interaction

The components interact with each other to provide a connected retail experience.

• The Customer Emotion Detection system collects behavioral and emotional data from customers in the store.

• This information is shared with the Smart Mirror Recommendation System to improve product suggestions.

• The Mobile AR/VR Recommendation System checks product availability through the Inventory Management system before displaying recommendations.

• Customer interaction data from the mirror is also shared with the Marketing Intelligence system to improve personalized promotions.

4.3 Scalability and Security Consideration

To support large retail environments, the system is designed with scalability and security in mind.

Scalability

• The system can support multiple smart mirrors across different store locations.

• Cloud-based data storage allows the system to process large amounts of customer interaction data.

• AI models can be updated and improved as more data is collected.

Security and Privacy

• Customer data will be anonymized to protect personal identity.

• The system will follow data protection regulations when collecting and storing customer information.

• Secure communication protocols will be used to protect data between system components.

5. User Requirements

5.1 Stakeholder Identification

The Mobile AR/VR Recommendation System involves several stakeholders who interact with or benefit from the system.

• Customers – People who use the smart mirror in the store to try products virtually and receive personalized recommendations.
• Store Managers – Retail managers who monitor customer interactions and use insights from the system to improve product placement and store performance.
• Retail Staff – Employees who assist customers and maintain the smart mirror system in the store.
• System Developers – Technical experts responsible for developing, maintaining, and improving the smart mirror system.

5.2 Functional Requirements

1. The system shall detect when a customer opens the mobile application and starts using the AR/VR virtual try-on interface.
2. The system shall allow customers to try products virtually using the mobile AR/VR interface.
3. The system shall analyze customer facial expressions and emotional responses in real time to understand their reactions to products.
4. The system shall analyze customer behavior, such as interaction time with a product, gestures, and which items they try virtually.
5. The system shall recommend suitable products based on customer emotions, features, preferences, and interaction behavior.
6. The system shall connect with the store inventory database to recommend only products that are available in real time.
7. The system shall display product details, such as price, color options, sizes, and availability, within the mobile AR/VR interface.
8. The system shall record customer interaction and emotional data to improve future recommendations while ensuring privacy.

5.3 **Non-Functional Requirements**

Security
• The system must protect customer data and ensure privacy.

Performance
• The system must provide fast product recommendations in real time.

Usability
• The smart mirror interface must be simple and easy for customers to use.

Reliability
• The system must work continuously during store hours without failures.

Scalability
• The system must support multiple smart mirrors in different retail stores.

6. COMMERCIALIZATION PLAN

- **Target Market and Customer Persona**

The target market for the Smart Retail Recommendation System includes retail businesses such as supermarkets, shopping malls, clothing stores, and large retail chains. The main users will be storing owners, store managers, and retail technology providers who want to improve customer experience, increase sales, and better understand customer behavior.

- **Value Proposition**

The system provides a smart and personalized shopping experience by combining technologies such as emotion detection, smart mirror recommendations, inventory management, and personalized marketing. This helps retailers understand customer needs in real time and improve product recommendations and sales.

- **Revenue Model**

Revenue can be generated through:

- Software subscription fees (monthly or yearly)
- System installation and setup services
- Maintenance and technical support services
- Retail data analytics services

- **Cost Estimation**

Main costs include:

- Software development for AI models and system platform
- Hardware costs such as cameras and smart mirrors
- Cloud storage and computing
- System maintenance and support

- **Pricing Strategy**

The system will use a tier-based pricing model:

- Basic Package – Basic recommendation features for small stores
- Standard Package – Includes emotion detection and smart mirror features
- Premium Package – Full system with advanced analytics and real-time assistance

- **Competitive Advantage**

The system offers:

- Integration of multiple smart retail technologies
- Real-time emotion and behaviour detection
- Personalized product recommendations
- Staff notifications for customer assistance

- **Intellectual Property Considerations**

The system can be protected through:

- Software copyright for the developed system
- Patents for unique technologies
- Trademark registration for the product name and brand.

7. BUDGET AND JUSTIFICATION

- **Hardware costs**

Table 2: Hardware Costs

Item	Estimated Costs
Laptop / Desktop PC	LKR 215,000

***Justification*:** Hardware resources are required to develop, train, and test the machine learning models.

- **Software Licenses**

Most tools used in this research are open-source.

EX: Python

TensorFlow / PyTorch

Scitkit-Learn

PostgreSQL / MongoDB

- **Cloud Services**

Table 3: Cloud Services Costs

Service	Cost
Cloud Computing for AI training	LKR 35000.00
Cloud storage	LKR 25000.00

***Justification*:** Cloud services enable scalable data storage and model deployment.

- **Human Effort Cost**

- Since this is an academic research project development is carried out by the student research team.
- Estimated academic effort value.

Table 4: Human Effort Cost

Role	Estimated Value
Research & Development	LKR 20000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) is used to identify tasks involved in developing the Smart Inventory and Stock Flow Optimization module of the proposed system. This module is designed to improve the accuracy of demand forecasting, optimize stock movement between stores and warehouses, and predict transportation costs using artificial intelligence techniques

Table 5: Work Breakdown Structure

Task ID	Task Name	Description
T1	Literature Review	Study existing research on demand forecasting, inventory optimization, and explainable AI
T2	Dataset Collection	Gather simulated retail sales datasets and contextual data such as weather and seasonal trends
T3	Data Preprocessing	Clean datasets, remove inconsistencies, and prepare structured data for model training
T4	Feature Engineering	Create features such as seasonal demand indicators, weather influence, and product sales patterns
T5	Demand Forecasting Model Development	Develop machine learning model to predict product demand
T6	Model Training & Evaluation	Train models and evaluate prediction accuracy
T7	Stock Flow Optimization Design	Design algorithm to identify stock shortages and excess stock
T8	Stock Transfer Recommendation System	Develop logic to recommend stock transfer between stores and warehouses

T9	Transport Cost Prediction Model	Develop AI model to estimate logistics cost for stock Transfers
T10	Explainable AI Integration	Implement explainable AI methods to provide reasoning behind predictions
T11	System Integration	Integrate forecasting, optimization and cost prediction modules
T12	Testing and Validation	Test System performance using simulated datasets
T13	Retail Analysis	Analyze prediction accuracy and optimization performance
T14	Documentation	Prepare research documentation and evaluation reports

9. Timeline Alignment

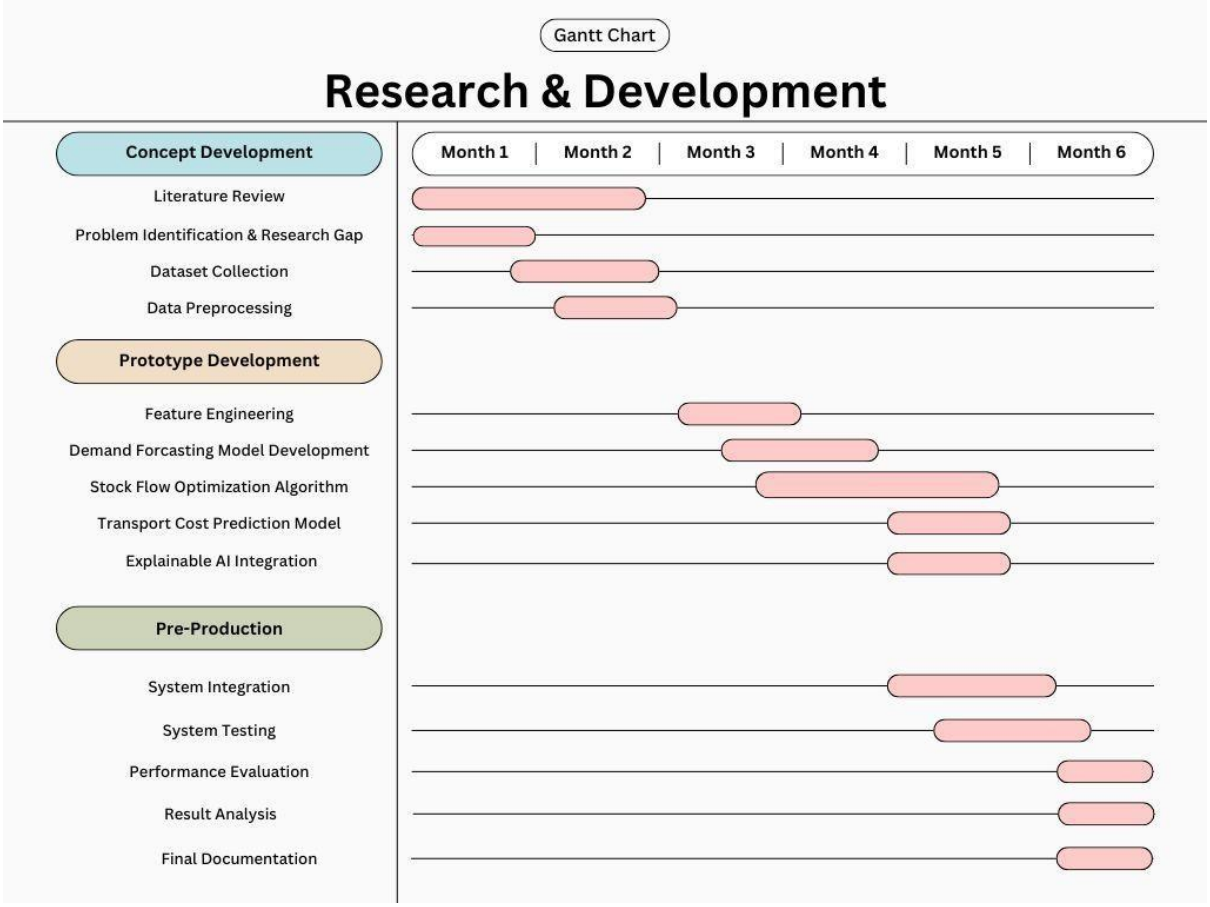
The development of the Smart Inventory and Stock Flow Optimization module will be conducted over a period of six months

Phase	**Activity**
Phase 1	Literature review and research planning
Phase 2	Dataset collection and preprocessing
Phase 3	Demand forecasting model development
Phase 4	Stock flow optimization and transport cost prediction
Phase 5	System testing and evaluation
Phase 6	Documentation and final research report

Balanced workload Distribution

The activities involved in the Smart Inventory and Stock Flow Optimization module are spread across different research phases, ensuring a balanced workload for the entire project timeline. The development process for this module consists of data preparation, developing the machine learning model, optimization algorithm implementation, and system evaluation.

10. GANTT CHART



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12. APPENDICES

12.1 **Appendix A - Survey Questions**

1. How often do you shop for clothing or fashion items?
2. Do you currently use mobile apps for shopping?
3. Have you ever used Augmented Reality (AR) features (such as virtual try-on) in a shopping app?
4. Have you ever used Virtual Reality (VR) for shopping or exploring stores?
5. How comfortable are you using new technologies like AR and VR in shopping?
6. Do you think AR can help you better understand how a product looks before buying?
7. Would you like to try clothes virtually using AR before making a purchase?
8. Do you think VR can improve your shopping experience by allowing you to explore virtual stores?
9. How important is it for you to see products in a realistic way before buying?
10. Do you trust product recommendations provided by mobile applications?

12.2 Appendix B – Ethical Considerations

This research follows ethical guidelines to ensure responsible data usage and system development.

• The research uses simulated retail data rather than confidential company data.

• No personal customer information will be collected or analyzed.

• The system focuses only on product inventory and sales patterns.

• Any survey responses collected will remain anonymous and used solely for academic purposes.

• The research does not involve sensitive personal data or harmful technologies

13.12. CONCLUSION

In conclusion, this proposal presents the development of a Smart Mirror Recommendation System, which is an important component of the Smart Retail Recommendation System. The aim of this component is to improve the in-store shopping experience by providing interactive and personalized product recommendations through a smart mirror interface.

The proposed system uses computer vision and artificial intelligence to analyze customer facial expressions, behavior, and interactions while they use the smart mirror. Based on this information and the store inventory data, the system recommends suitable products that match customer preferences. This approach helps customers make better shopping decisions and also helps retailers understand customer interests more effectively.

Overall, this system introduces a more intelligent and personalized retail environment. It moves beyond traditional smart mirrors that only provide virtual try-on features by adding smart recommendations and real-time product availability. This innovation can improve customer satisfaction, increase sales opportunities, and support the future development of AI-based smart retail technologies.